

CAMDEN DA SILVA

Engineering Portfolio

BASc Nanotechnology Engineering, Honours Co-op · University of Waterloo
Nanofabrication & Nanoelectronics specialization · Expected 2028
cdasilva@uwaterloo.ca · +1 (705) 351-3535 · LinkedIn

Why this work

Most of what follows is hardware I specified, built and then had to debug. A spin-coating platform that drives a syringe pump and a closed-loop BLDC head in step, where the firmware was setting speed from a motor rating that turned out to be wrong by as much as 4,700 RPM. A resonant thin-film sensor whose analog front end was simulated as an equivalent circuit before any topology was chosen, and whose thermal design was settled by comparing five mounting configurations two independent ways. The remaining pages are semiconductor devices built by hand and the metrology used to decide whether to believe them. The thread running through all of it: in three of these projects the first answer the instrument gave was wrong, and the real work was finding out why.

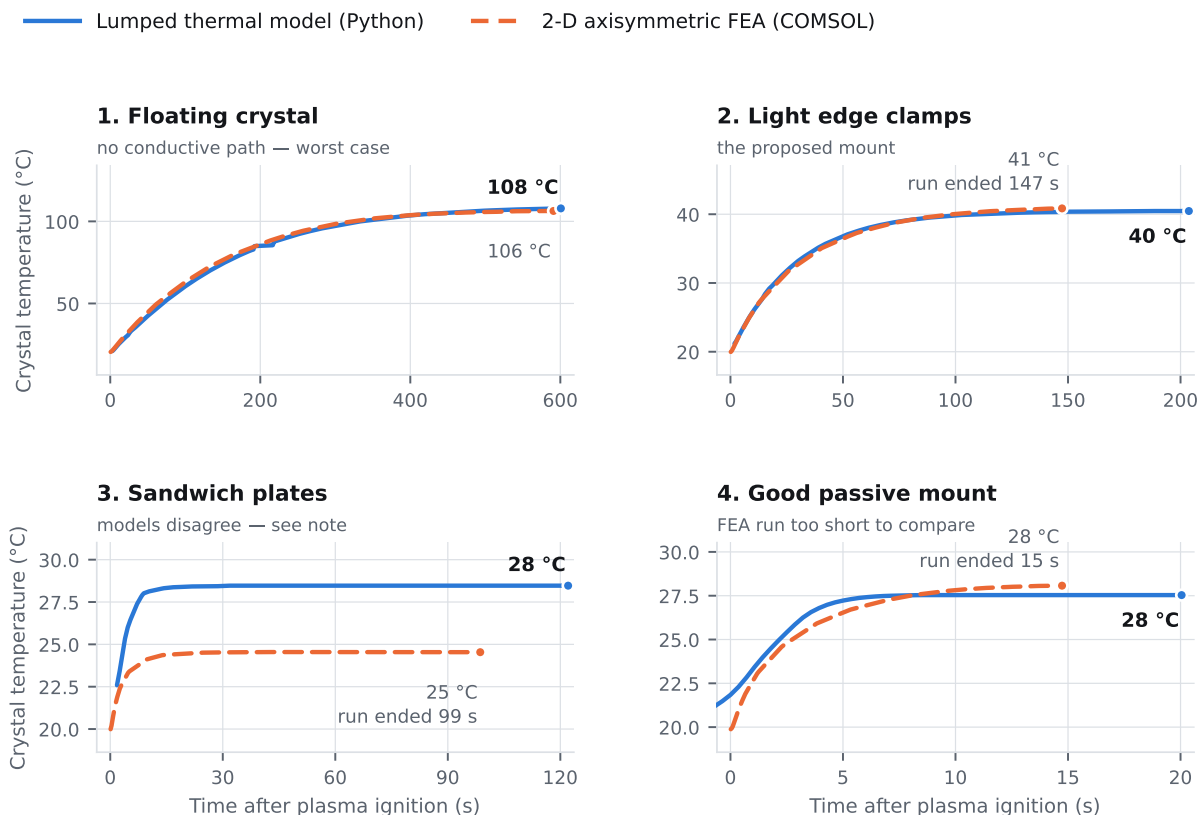
Designing Out Thermal Drift in a Resonant Thin-Film Sensor

Waterloo Hacker Fab — DC magnetron sputtering system · Spring 2026 – present

Problem The sputtering system needs to know how thick a film is while it is growing. A quartz crystal microbalance infers deposited mass from a shift in the crystal's resonant frequency, which means anything else that changes the resonator — including radiant heat from the sputter head — is indistinguishable from deposition. A QCM that drifts does not fail visibly; it returns a confident wrong number. The mount had to be chosen before the sensor was built, because the mount is what sets how much of that heat reaches the crystal.

Approach I built a transient lumped-parameter thermal model in Python covering five candidate mounting configurations, from a bare floating crystal to a water-cooled holder. Incident flux from the plasma was taken as 0.03 W/cm^2 , and effective emissivity — by far the least certain input — was swept from 0.05 to 0.75 rather than assumed. I then built an independent 2-D axisymmetric finite-element model in COMSOL from the SolidWorks enclosure geometry and ran four of the configurations again, as a check on the lumped model rather than a repeat of it.

Result A floating crystal settles near 108°C ; light edge clamps hold it at 40°C for a small conductive penalty, which is the configuration selected for the build. The two models agree to 0.4°C on the clamped case and 1.6°C on the floating case. They disagree by 4°C on the sandwich-plate case, and the passive-mount FEA run was stopped at 15 s before it reached steady state — both are open, and both are shown below rather than left out. A smaller correction mattered too: a 6 MHz AT-cut crystal is 0.277 mm thick, not the 0.33 mm commonly quoted, and the thermal mass follows from that.



Crystal temperature after plasma ignition for four candidate mounts, from two independent models. Note the differing time axes: the mounts differ in settling time by more than an order of magnitude.

Tools Python · COMSOL Multiphysics · SolidWorks · vacuum system build-up

Thermal Drift, Continued — What the Design Is Actually Sensitive To

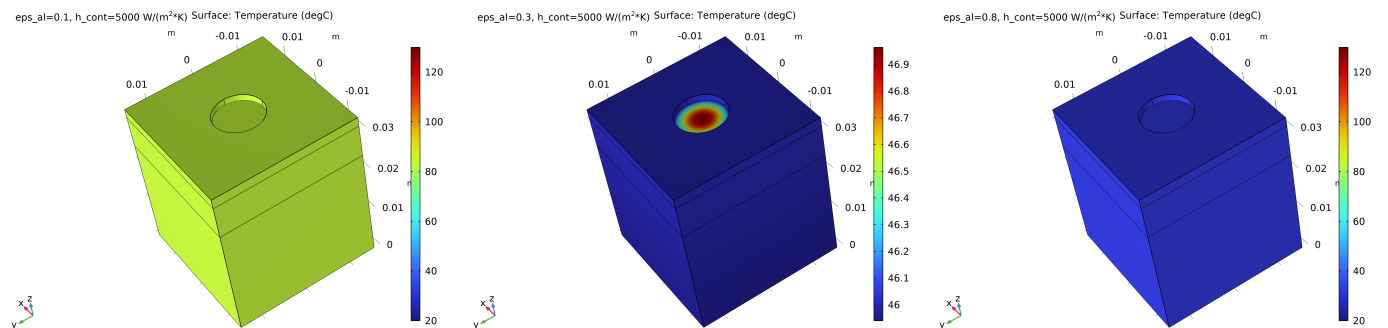
Waterloo Hacker Fab — three-dimensional COMSOL parametric study · September 2026

Question Choosing a mount answered one question and raised another: of all the numbers that went into that model, which ones does the answer actually depend on? Two of them are still open to whoever builds the thing. The surface finish of the aluminium housing sets how readily it radiates heat away, and the conductance across the crystal-to-holder interface is the least certain input in the model, being a fitted quantity rather than a datasheet value.

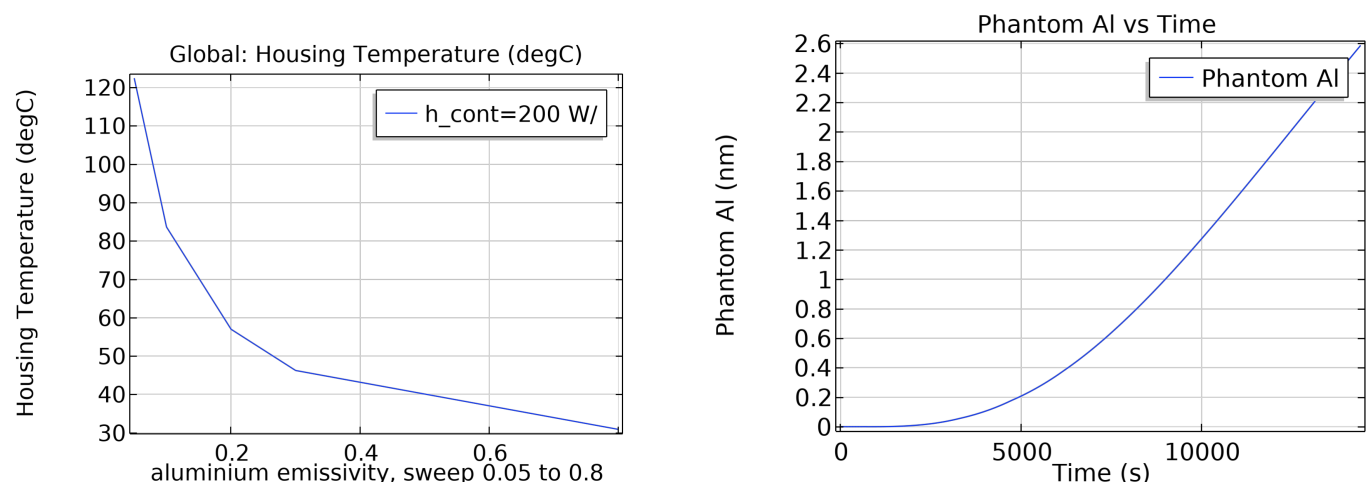
Approach I rebuilt the enclosure as a full three-dimensional COMSOL model from the SolidWorks geometry, in place of the earlier axisymmetric slice, and swept both inputs: aluminium emissivity from 0.05 to 0.8, which spans bare polished aluminium at one end and an anodised finish at the other, and contact conductance from 200 to 5,000 W/(m²·K).

Result Emissivity decides the design and nothing else comes close. Housing temperature falls from 122 °C at $\varepsilon = 0.05$ to 83 °C at 0.1, 46 °C at 0.3, and 31 °C at 0.8: a 91 °C span bought with a surface treatment. Most of it arrives early, since reaching $\varepsilon = 0.3$ removes 76 °C and the entire remaining range is worth another 15. Contact conductance, the input I was least sure of, turned out to matter least. A factor of twenty-five increase moves the housing by a few degrees at $\varepsilon = 0.1$ and by essentially nothing at 0.3, which is a useful thing to know before building, because it means the design does not rest on controlling an interface that is hard to control. The crystal tracks the housing closely, sitting about 0.9 °C above it, so housing temperature is the quantity worth designing against.

In measurement units Converted into what the instrument would actually report, the drift stays below 0.3 nm of apparent aluminium through the first 5,000 s and reaches roughly 2.6 nm over a four-hour run. Short depositions are not thermally limited at all; only long ones need the compensation path built. The instruction this study produces is a finish specification rather than a mechanism: anodise the housing.



Steady-state housing temperature at three aluminium emissivities, contact conductance 5,000 W/(m²·K). Left and right share one 20–130 °C scale; the centre frame is auto-scaled across 0.9 °C to show the crystal pocket above the housing.



Left: housing temperature against emissivity. **Right:** the drift as apparent aluminium thickness, from heat alone.

Tools COMSOL Multiphysics (three-dimensional, parametric sweep) · SolidWorks · Python

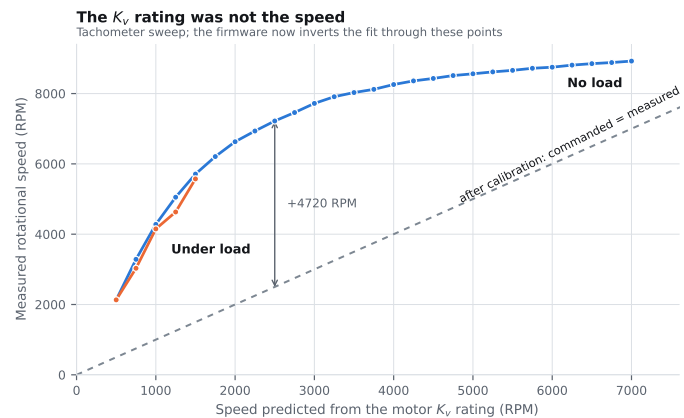
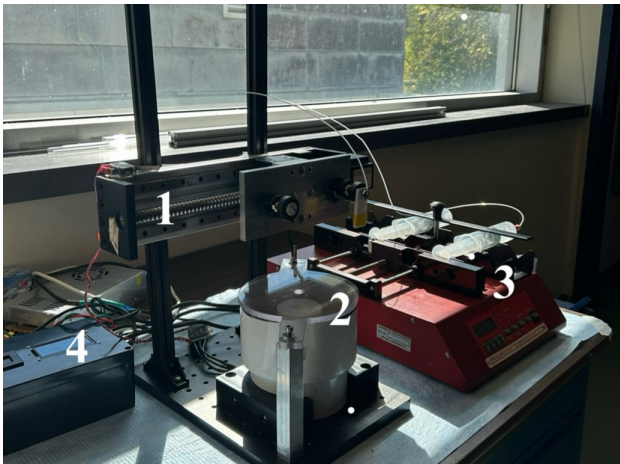
Automated Layer-by-Layer Deposition Platform

McGill University, Soft Matter Lab · Jan–Apr 2025; Sept 2025–Apr 2026

Problem The group's layer-by-layer deposition was done by hand, so repeatability was limited by whoever was doing it. It needed an instrument that produced the same film twice.

Approach I specified and assembled the platform from components, wrote the microcontroller firmware, and coordinated a syringe pump with a closed-loop BLDC drive so that dispensing and rotation stayed synchronised. The firmware initially set speed from the motor's K_v rating, so before running any films I checked it: I swept the full range against a tachometer and recorded actual rotational speed, loaded and unloaded.

Result The K_v rating was not the speed. Measured rotation ran between 1,630 and 4,720 RPM above what the rating predicted, and the error was not a constant offset or a constant factor — it grew toward mid-range and then closed again, so nothing simple would have corrected it. I fitted the measured curve and inverted that fit in firmware, so a commanded 5,000 RPM now turns at 5,000 RPM and every process parameter in the recipe is a real speed rather than a nominal one. The rig's measured ceiling is roughly 8,900 RPM unloaded. With the drive calibrated I established the process window by structured sweeps over speed, dispense volume and drying: an earlier recipe left finger-like drying patterns across the film, and the settled recipe removed them. Films came out at roughly 11–18 nm per bilayer, measured by stylus profilometry over 30- and 100-layer stacks and consistent with published values for the same system.



Left: the platform — (1) linear stage, (2) spin-coating head on the BLDC drive, (3) syringe pump, (4) control electronics. **Right:** speed predicted from the K_v rating against measured speed. Inverting a fit through these points is what puts the drive back on the dashed line, where the commanded value is the actual one.

Tools Embedded C/C++ · closed-loop motor control · syringe-pump actuation · stylus profilometry · MATLAB analysis

Cross-Validating a Thin-Film Thickness Measurement

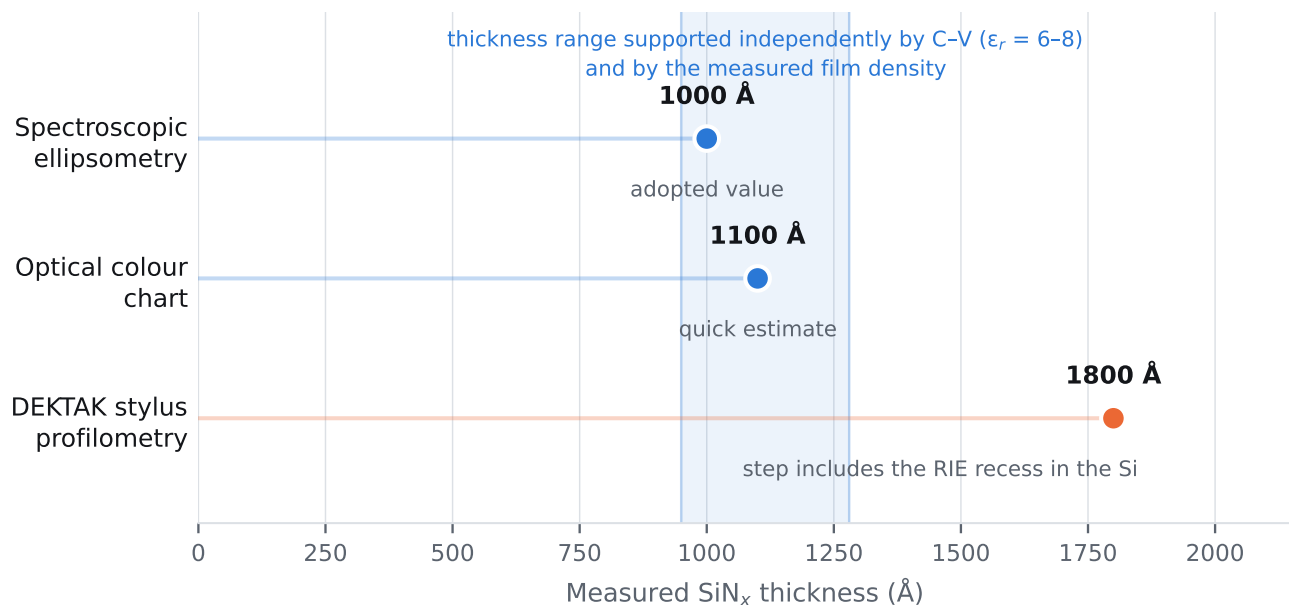
Course project — NE 343 Microfabrication & Thin-Film Technology · Spring 2026

Problem I measured the same PECVD silicon nitride film three ways and got three answers, spanning nearly a factor of two. Any one of them, reported alone, would have been taken as the thickness — and everything downstream, including the dielectric constant of the film, scales directly with whichever one you believe.

Approach Optical colour chart, DEKTAK stylus profilometry and spectroscopic ellipsometry disagreed. Rather than average them, I looked for evidence that did not come from any of the three. Two independent checks were available: the film's mass, weighed on the wafer before and after deposition, and the maximum capacitance of MIS devices built on the same film.

Result The ellipsometric value of 1000 Å is the one the independent evidence supports. At that thickness the measured 1.2 mg of film over the wafer area gives a density of 2.63 g/cm³, inside the 2.5–3.1 g/cm³ range reported for PECVD nitride; at 1800 Å it gives 1.46 g/cm³, below any published value for the material. The electrical check agrees: 1000 Å yields $\epsilon_r = 6.2$, within the usual 6–9 for PECVD SiN_x, while 1800 Å would require 11.2. The profilometer was not wrong so much as answering a different question — RIE has finite nitride-to-silicon selectivity, so the exposed silicon was etched below the original surface and the measured step contains both the film and the recess beneath it. The useful outcome is not the number but the rule: a stylus measures a step, not a film, and the two are only the same when the etch stops where you think it did.

Three methods, three answers on the same film



The three direct measurements against the thickness range that the mass and capacitance data independently support.

Tools Spectroscopic ellipsometry (Woollam alpha-SE) · DEKTAK stylus profilometry · reactive ion etch · gravimetric analysis

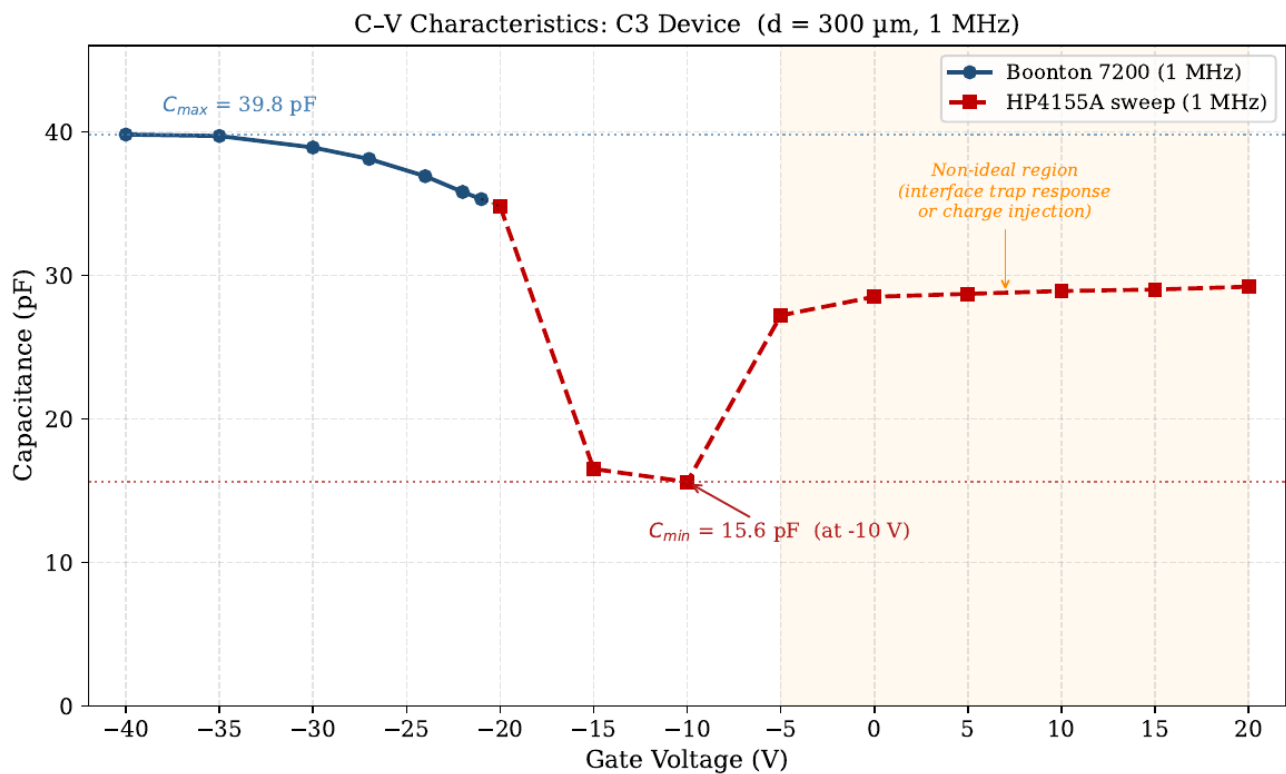
MIS Capacitor — Full Fabrication Flow and Electrical Test

Course project — NE 343 Microfabrication & Thin-Film Technology · Spring 2026

Problem Build an Al/SiN_x/p-Si metal–insulator–semiconductor capacitor from a bare wafer across three clean-room sessions, and determine from its electrical behaviour what the dielectric and the interface are actually doing.

Approach Complete flow: RCA clean, PECVD silicon nitride, contact photolithography, reactive ion etch, DC magnetron sputtered aluminium, wet etch. Then C–V at 1 MHz and I–V on nine devices in a cross-pattern, across five contact diameters and five wafer positions, so that area effects and spatial variation could be separated.

Result $C_{\max}$ against contact area is linear to $R^2 \approx 1.00$, giving $\epsilon_r = 6.2$. From $C_{\max} = 39.8 \text{ pF}$ and $C_{\min} = 15.6 \text{ pF}$ the substrate doping works out to $N_a \approx 1.1 \times 10^{16} \text{ cm}^{-3}$, which lands inside the wafer's specified resistivity range — an independent confirmation that the extraction is sound. The interesting part is where the device is not ideal. Flatband sits near -20 V against an ideal -0.68 V , a shift of -19.3 V corresponding to a positive fixed charge density of $6.8 \times 10^{12} \text{ cm}^{-2}$ in the nitride, which is what SiH₄/NH₃ PECVD film is expected to carry. Capacitance stayed within $\pm 3\%$ across the wafer and leakage below 10^{-10} A out to 20 V with no breakdown: a film with a lot of fixed charge, but a spatially uniform and electrically robust one. What stayed with me was how tightly the steps couple — what you accept at develop, you pay for at etch, and then again at extraction.



C–V characteristic of the 300 μm device at 1 MHz. The elevated capacitance through inversion, well above $C_{\min}$, is interface states fast enough to follow the 1 MHz signal.

Tools PECVD · DC magnetron sputtering · photolithography · reactive ion etch · C–V and I–V characterisation · four-point probe

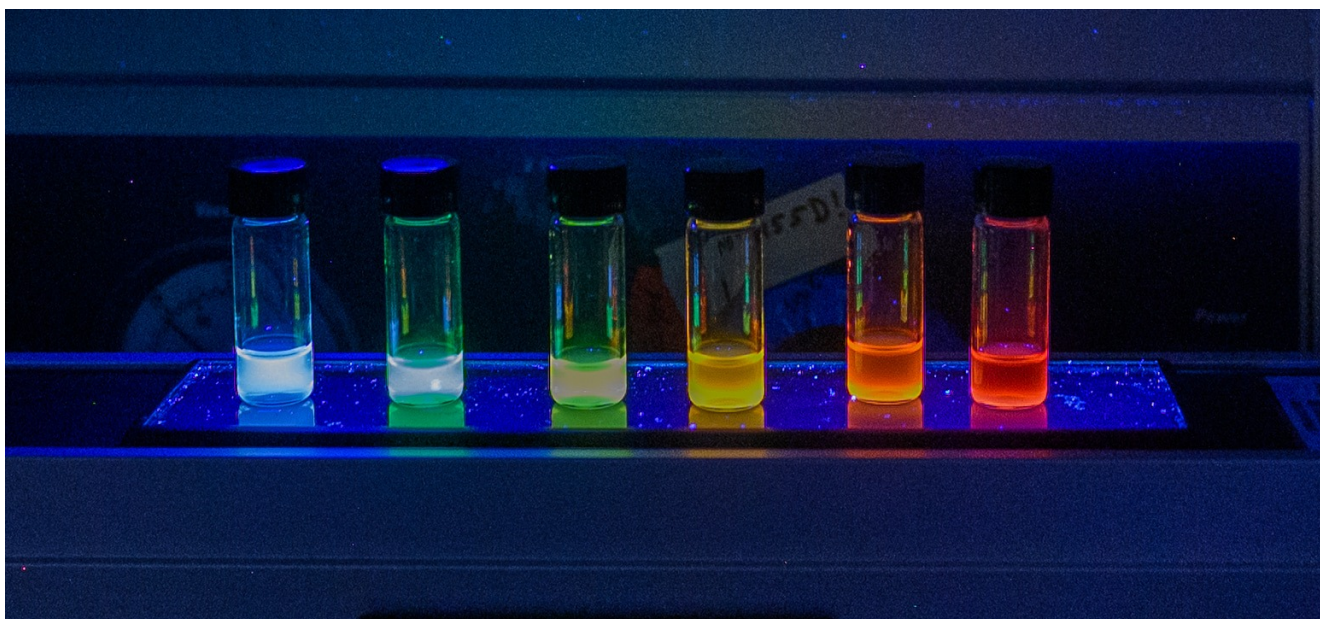
Quantum-Dot LED from Synthesised Material

Course project — Nanomaterials Synthesis Laboratory · Spring 2026

Problem Take a nanomaterial from precursors to a device that emits light, and show that the structural measurement and the optical behaviour agree.

Approach Synthesised CdSe quantum dots, SPIONs, and europium- and dysprosium-doped phosphors, then built a spin-cast quantum-dot LED from the CdSe. Emission was tuned by controlling growth time alone, and I correlated image-derived size distributions against measured emission using the Brus model, with magnetic response treated through Néel and Brownian relaxation.

Result Closing the loop from particle size to device output is the part I care about — a spectrum on its own does not tell you whether you understand the material.



The CdSe size series under UV excitation. Every vial is the same chemistry; only growth time differs, and the emission moves across the visible with particle size.

Tools Colloidal synthesis · spin coating · photoluminescence · UV-Vis · PXRD · SEM